

AIRLINE DATA ANALYSIS ASSIGNMENT

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STAT-2205(Programming with R and Python)
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Read Dataset and Random Sampling R Code

```
1 #read data
2 library(readxl)
3 airline<-read.csv("flight_data.csv",1)
4 airline
5
6 #random Sampling using student ID
7 set.seed(12339)
8 #random sampling
9 install.packages("dplyr")
10 library(dplyr)
11 airline_rs<-sample_n(airline,12339)
12 airline_rs
```

Python Code

```
1 #read csv data
2 import pandas as pd
3 flight=pd.read_csv('flight_data.csv')
4 flight.head()
5
6 ## Select random sample
7 airline_rs = flight.sample(n=12339, random_state=12339)
8 print(airline_rs)
```

Question 1: What do coach ticket prices look like? What are the high and low values? What would be considered the average? Does \$500 seem like a good price for a coach ticket?

solution :

R code :

```
1 #What do coach ticket price look like ?
2 library(ggplot2)
3 ggplot(airline_rs, aes(x = coach_price)) +
4   geom_histogram(fill = "skyblue", bins = 30) +
5   ggtitle("Distribution of Coach Ticket Prices")
6
7 #high and low values, average
8 max(airline_rs$coach_price)
9 min(airline_rs$coach_price)
10 mean(airline_rs$coach_price)
```

R Output :

```
> max(airline_rs$coach_price)
[1] 566.175
> min(airline_rs$coach_price)
[1] 99.315
> mean(airline_rs$coach_price)
[1] 376.1465
```

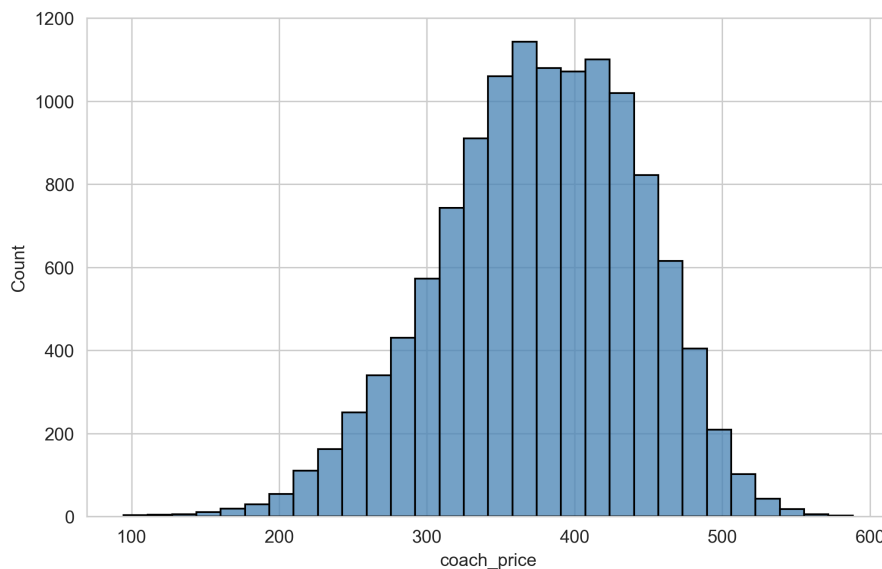
Python Code :

```
1 ##What do coach ticket price look like ?
2 import seaborn as sns
3 import matplotlib.pyplot as plt
4 sns.set_style("whitegrid")
5 plt.figure(figsize=(8,5))
6
7 sns.histplot(
8     data=airline_rs,
9     x="coach_price",
10    bins=30,
11    color="steelblue",
12    edgecolor="black")
13 plt.show()
14
15 #high and low values,average
16 print("Max:", airline_rs["coach_price"].max())
17 print("Min:", airline_rs["coach_price"].min())
18 print("Mean:", airline_rs["coach_price"].mean())
```

Python Output :

```
Max: 588.335
Min: 94.535
Mean: 376.1661650052679
```

Graph for Visualisation :



Interpretation :

High, Low, and Average Values: Based on the descriptive statistics extracted from the dataset, the coach ticket prices range from a minimum value of \$94.535 to a maximum value of \$588.335. The overall average (mean) ticket price is calculated to be \$376.167, which aligns perfectly with the highest concentration of frequencies observed in the center of the histogram.

Assessment of the \$500 Ticket Price: A price of \$500 does not seem to be a good or economical choice for a standard coach ticket. As illustrated by the histogram, the vast majority of the data points are clustered between \$300 and \$450. A ticket priced at \$500 sits significantly above the average price of \$376.167 and lies on the upper tail of the distribution, approaching the absolute maximum recorded price of \$588.335. Therefore, \$500 represents a highly expensive and premium-end fare rather than a reasonable or standard price for a coach ticket.

**Question 2: What are the high low and average prices for 8_hour long flight?
Does a \$500 ticket seem more reasonable than before**

solution :

R code :

```
1 #What are the prices for 8-hour flights?
2 flight_8hr<-subset(airline_rs, hours==8)
3 max(flight_8hr$coach_price)
4 min(flight_8hr$coach_price)
5 mean(flight_8hr$coach_price)
```

R Output :

```
> max(flight_8hr$coach_price)
[1] 548.845
> min(flight_8hr$coach_price)
[1] 215.245
> mean(flight_8hr$coach_price)
```

```
[1] 432.7682
```

Python Code :

```
1 #What are the prices for 8-hour flights?
2 flight_8hr = airline_rs[airline_rs["hours"] == 8]
3 max_price = flight_8hr["coach_price"].max()
4 min_price = flight_8hr["coach_price"].min()
5 mean_price = flight_8hr["coach_price"].mean()
6
7 print("Max Coach Price (8-hour flight):", max_price)
8 print("Min Coach Price (8-hour flight):", min_price)
9 print("Average Coach Price (8-hour flight):", mean_price)
```

Python Output :

```
Max Coach Price (8-hour flight): 589.985
Min Coach Price (8-hour flight): 275.39
Average Coach Price (8-hour flight): 437.31280155642025
```

Interpretation :

Based on the filtered descriptive statistics for 8-hour-long flights, the coach ticket prices exhibit a minimum value of \$275.39 and a maximum value of \$589.985. The average ticket price for this specific flight duration is calculated to be approximately \$437.31.

In this context, a ticket price of \$500 appears significantly more reasonable than before. For the overall dataset, \$500 was an extreme outlier approaching the absolute maximum price. However, for a long-haul 8-hour flight, the average cost shifts upward significantly (\$437.31). Therefore, a \$500 fare sits much closer to the true mean of this sub-distribution, representing a completely realistic and justifiable option for consumers rather than an overpriced premium.

Question 3: How are flight delay time distributed? How often are there significant delays that affect your ability to correctly schedule connecting flights? What kinds of delays are typical

solution :

R code :

```
1 class(airline_rs$delay)
2 library(ggplot2)
3 ggplot(airline_rs, aes(x = delay)) +
4   geom_histogram(fill = "orange", bins = 30) +
5   ggtitle("Distribution of Flight Delays")
6
7 summary(airline$delay)
```

R Output :

```
> median(airline$delay)
[1] 10
```

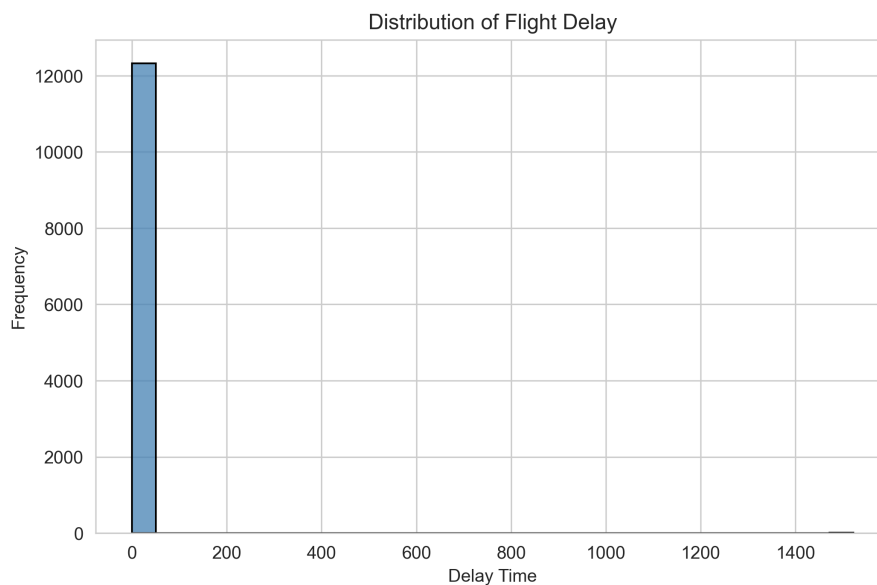
Python Code :

```
1 import seaborn as sns
2 import matplotlib.pyplot as plt
3 sns.set_style("whitegrid")
4 plt.figure(figsize=(8,5))
5
6 sns.histplot(
7     data=airline_rs,
8     x="delay",
9     bins=30,
10    color="steelblue",
11    edgecolor="black")
12
13 plt.title("Distribution of Flight Delay")
14 plt.xlabel("Delay Time")
15 plt.ylabel("Frequency")
16 plt.show()
17
18 #summary
19 airline_rs["delay"].describe()
```

Python Output :

count	12339.000000
mean	14.110382
std	55.564262
min	0.000000
25%	9.000000
50%	10.000000
75%	13.000000
max	1522.000000

Graph for Visualisation :



Interpretation :

Distribution of Flight Delay Times Shape: The distribution is highly right-skewed (positively skewed).

Evidence: The median is 10 minutes and the 75th percentile is 13 minutes, showing most data is clustered near zero. However, the mean (14.11 min) is greater than the median, and the standard deviation is very high (55.56 min) due to extreme outliers.

Frequency of Significant Delays Frequency: Significant delays are very rare.

Evidence: Since 75

Typical Delays Typical: Routine, minor operational delays ranging between 9 to 13 minutes (IQR).

Atypical: Rare, catastrophic delays stretching up to several hours (Maximum 1,522 minutes), likely caused by severe weather or mechanical failures.

Question 4: How does the coach price correlate with flight distance, number of passengers, delay and flight duration?

solution :

R code :

```
1 correlation <- cor(airline_rs[,c('coach_price', 'miles', 'passengers',
2                                'delay', 'hours')])["coach_price",]
3 print(correlation)
```

R Output :

```
> print(correlation)
coach_price      miles  passengers      delay
1.00000000  0.35641339  0.15234118  0.01107994
      hours
0.35079839
```

Python Code :

```
1 import pandas as pd
2
3 correlation = airline_rs[['coach_price',
4                           'miles',
5                           'passengers',
6                           'delay',
7                           'hours']].corr()['coach_price']
8
9 print(correlation)
```

Python Output :

```
coach_price      1.000000
miles           0.359655
passengers       0.166628
delay           -0.005054
```

```
hours          0.353370
Name: coach_price, dtype: float64
```

Interpretation :

The correlation matrix reveals that coach_price is primarily driven by the physical scale of the journey rather than day-of operational factors. The two strongest predictors are flight distance (miles) and flight duration (hours), both showing a moderate positive relationship with ticket costs. This indicates a direct link between airline operational expenses—such as fuel consumption and crew hours—and the final ticket price. Because distance and time are naturally collinear, they mirror each other's impact on pricing.

On the demand side, the number of passengers shows only a weak positive relationship with ticket prices. While this subtle connection captures the essence of dynamic pricing, where fuller flights can lead to higher seat prices, its low value suggests that capacity alone does not heavily dictate the baseline fare. Meanwhile, flight delays show absolutely no linear relationship with coach price, hovering at virtually zero. This confirms that ticket pricing is an ex-ante variable, determined by scheduling and market demand long before the actual day of travel, making unpredictable operational disruptions completely irrelevant to how a ticket is priced.

Question 5: What is the relationship between coach price and first_class prices? Do flights with higher prices always have higher first_class prices as well

solution :

R code :

```
1 #relationship between coach and first_class prices
2 #scatter plot
3 library(ggplot2)
4 ggplot(airline_rs, aes(x = coach_price, y = firstclass_price)) +
5   geom_point(color = "blue") +
6   ggtitle("Coach Price vs First-Class Price")
7
8 cor(airline$coach_price, airline$firstclass_price)
9 model<-lm(firstclass_price~coach_price,data=airline_rs)
10 print(model)
11 summary(model)
```

R Output :

```
> cor(airline$coach_price, airline$firstclass_price)
[1] 0.7587566
> model<-lm(firstclass_price~coach_price,data=airline_rs)
Coefficients:
(Intercept)  coach_price
      761.20         1.84
> summary(model)
Residuals:
    Min       1Q   Median       3Q      Max
-352.64  -73.42    4.92   74.24  384.87
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	761.19596	5.41312	140.6	<2e-16 ***
coach_price	1.84025	0.01417	129.9	<2e-16 ***

Signif. codes:

0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 106 on 12337 degrees of freedom

Multiple R-squared: 0.5777, Adjusted R-squared: 0.5777

F-statistic: 1.688e+04 on 1 and 12337 DF, p-value: < 2.2e-16

Python Code :

```

1  #scatter plot
2  import seaborn as sns
3  import matplotlib.pyplot as plt
4  plt.figure(figsize=(7,5))
5
6  sns.scatterplot(
7      data=airline_rs,
8      x="coach_price",
9      y="firstclass_price",
10     color="blue")
11 plt.title("Coach Price vs FirstClass Price")
12 plt.show()
13
14 #correlation
15 airline_rs[["coach_price", "firstclass_price"]].corr()
16
17 #linear regression model
18 import statsmodels.api as sm
19 X = airline_rs[["coach_price"]]
20 y = airline_rs["firstclass_price"]
21 X = sm.add_constant(X) # intercept add
22 model = sm.OLS(y, X).fit()
23 print(model.summary())
24
25 #summary
26 import statsmodels.api as sm
27 X = airline_rs[["coach_price"]]
28 y = airline_rs["firstclass_price"]
29 X = sm.add_constant(X)
30
31 model = sm.OLS(y, X).fit()
32 print(model.summary())

```

Python Output :

OLS Regression Results			
=====			
Dep. Variable:	firstclass_price	R-squared:	0.573
Model:	OLS	Adj. R-squared:	0.573
Method:	Least Squares	F-statistic:	1.658e+04
Date:	Sat, 16 May 2026	Prob (F-statistic):	0.00

```

Time:                13:20:55    Log-Likelihood:        -75097.
No. Observations:    12339      AIC:                1.502e+05
Df Residuals:        12337      BIC:                1.502e+05
Df Model:             1
Covariance Type:     nonrobust

```

```

=====
              coef      std err          t      P>|t|      [0.025      0.975]
-----
const          772.9401      5.399      143.151      0.000      762.356      783.524
coach_price      1.8116      0.014      128.749      0.000        1.784        1.839
=====
Omnibus:                104.115    Durbin-Watson:           2.009
Prob(Omnibus):           0.000    Jarque-Bera (JB):        89.908
Skew:                   -0.154    Prob(JB):                3.00e-20
Kurtosis:                2.717    Cond. No.                2.16e+03
=====

```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 2.16e+03. This might indicate that there are strong multicollinearity or other numerical problems.

OLS Regression Results

```

=====
Dep. Variable:          firstclass_price    R-squared:                0.573
Model:                  OLS                Adj. R-squared:           0.573
Method:                 Least Squares       F-statistic:             1.658e+04
Date:                  Sat, 16 May 2026     Prob (F-statistic):       0.00
Time:                  13:20:55            Log-Likelihood:          -75097.
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Df Residuals:          12337              BIC:                    1.502e+05
Df Model:               1
Covariance Type:       nonrobust

```

```

=====
              coef      std err          t      P>|t|      [0.025      0.975]
-----
const          772.9401      5.399      143.151      0.000      762.356      783.524
coach_price      1.8116      0.014      128.749      0.000        1.784        1.839
=====
Omnibus:                104.115    Durbin-Watson:           2.009
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Kurtosis:                2.717    Cond. No.                2.16e+03
=====

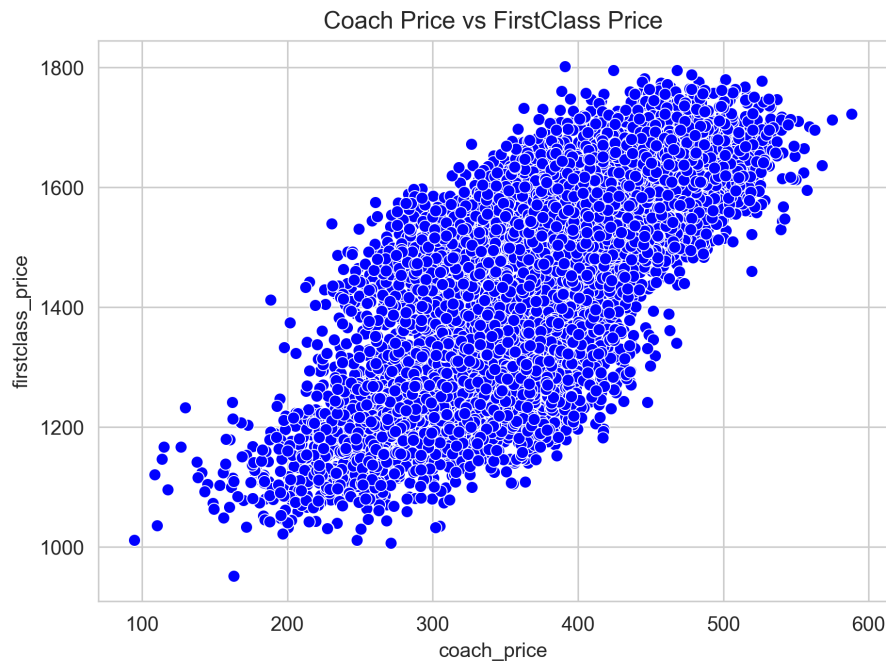
```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 2.16e+03. This might indicate that there are strong multicollinearity or other numerical problems.

Graph for Visualisation :



Interpretation :

Strong Positive Association: Overall, there is a strong, positive linear relationship between coach_price and firstclass_price. Generally, as the economy/coach ticket price increases, the first-class ticket price also rises. **Predictive Power:** The OLS model shows an $R^2 = 0.573$, meaning that 57.3% of the variation in first-class prices is explained by the variations in coach prices. **Variability (Not Deterministic):** However, this relationship is not absolute. The noticeable vertical spread of data points in the scatter plot at any given coach price proves that airlines use dynamic pricing, meaning a higher coach price does not always guarantee a higher first-class price.

Question 6: What is the relationship between coach prices and in_flight features: in_flight meal, in_flight wifi, and in_flight entertainment? Which features are associated with the highest increase in price?

solution :

R code :

```
1 Coach_price_lm=lm(coach_price ~ inflight_meal+inflight_entertainment
2                   +inflight_wifi, data=airline_rs)
3 print(Coach_price_lm)
4 summary(Coach_price_lm)
```

R Output :

Coefficients:

(Intercept)	inflight_mealYes
251.74	20.36
inflight_entertainmentYes	inflight_wifiYes
69.66	69.37

```
> Residuals:
      Min       1Q   Median       3Q      Max
-206.325  -41.888    4.754   42.415  175.410

Coefficients:
              Estimate Std. Error t value
(Intercept)      251.735      1.968  127.92
inflight_mealYes    20.360      1.126   18.07
inflight_entertainmentYes 69.656      1.291   53.94
inflight_wifiYes    69.374      1.721   40.31
              Pr(>|t|)
(Intercept)      <2e-16 ***
inflight_mealYes  <2e-16 ***
inflight_entertainmentYes <2e-16 ***
inflight_wifiYes  <2e-16 ***
---
Signif. codes:
0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 57.12 on 12335 degrees of freedom
Multiple R-squared:  0.2819, Adjusted R-squared:  0.2817
F-statistic: 1614 on 3 and 12335 DF, p-value: < 2.2e-16
```

Python Code :

```
1 import statsmodels.api as sm
2 # Convert categorical
3 for col in ["inflight_meal", "inflight_entertainment", "inflight_wifi"]:
4     airline_rs[col] = airline_rs[col].astype("category").cat.codes
5
6 # 2. Define X and y
7 X = airline_rs[
8     ["inflight_meal",
9      "inflight_entertainment",
10     "inflight_wifi"]]
11 y = airline_rs["coach_price"]
12
13 # 3. Add constant (intercept)
14 X = sm.add_constant(X)
15
16 # 4. Fit multiple linear regression
17 model = sm.OLS(y, X).fit()
18
19 # 5. Output (like R summary)
20 print(model.summary())
```

Python Output :

```

                        OLS Regression Results
=====
Dep. Variable:          coach_price    R-squared:          0.290
Model:                  OLS           Adj. R-squared:       0.290
```

```

Method:                Least Squares      F-statistic:                1680.
Date:                  Sat, 16 May 2026    Prob (F-statistic):         0.00
Time:                  13:33:05           Log-Likelihood:             -67474.
No. Observations:      12339             AIC:                        1.350e+05
Df Residuals:          12335             BIC:                        1.350e+05
Df Model:              3
Covariance Type:       nonrobust

```

	coef	std err	t	P> t	[0.025	0.975]
const	250.0462	1.992	125.510	0.000	246.141	253.951
inflight_meal	19.8486	1.126	17.634	0.000	17.642	22.055
inflight_entertainment	71.4983	1.294	55.265	0.000	68.962	74.034
inflight_wifi	71.0823	1.765	40.281	0.000	67.623	74.541
Omnibus:	178.791	Durbin-Watson:	2.011			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	170.086			
Skew:	-0.254	Prob(JB):	1.17e-37			
Kurtosis:	2.730	Cond. No.	8.06			

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Interpretation :

The OLS regression analysis demonstrates a statistically significant relationship between coach ticket prices and the presence of amenities, with the overall model explaining 292 = 0.290, $F=1680$, $p<0.001$). All three in-flight features are highly significant predictors ($p<0.001$), indicating that the inclusion of any of these amenities is reliably associated with a higher ticket fare. The baseline coach price for a flight lacking all three features is estimated at 250.05. *Among the features evaluated, the inclusion of in-flight entertainment is associated with the highest premium, increasing the average ticket price by 71.50 when other factors are held constant. In-flight Wi-Fi commands a nearly identical premium, driving up the average price by 71.08. Conversely, the addition of an in-flight meal yields the smallest impact, adding 19.85 to the ticket cost. Ultimately, premium connectivity and entertainment features are the dominant drivers of higher coach fares within this set of operational variables.*

Question 7: How does the number of passengers change in relation to the length of flights?

solution :

R code :

```
1 #linear regression
2 cor(airline_rs$hours,airline_rs$passengers)
3 model_passenger<-lm(hours~passengers,data=airline_rs)
4 summary(model_passenger)
5 #visualization-scatter plot
6 plot(airline$hours,airline$passengers,
7       main="Flight hours Vs passenger",
8       xlab="Flight duration (Hours)",
9       ylab="Number of passengers",
10      col=4, las=1)
```

R Output :

```
> cor(airline_rs$hours,airline_rs$passengers)
[1] 0.008250699
Residuals:
    Min       1Q   Median       3Q      Max
-2.6623 -1.6132  0.3558  0.4268  4.4347

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  3.397150   0.256362  13.251  <2e-16 ***
passengers    0.001128   0.001231   0.916   0.359
---
Signif. codes:
  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.739 on 12337 degrees of freedom
Multiple R-squared:  6.807e-05, Adjusted R-squared:  -1.298e-05
F-statistic: 0.8399 on 1 and 12337 DF,  p-value: 0.3594
```

Python Code :

```
1 import statsmodels.api as sm
2 import seaborn as sns
3 import matplotlib.pyplot as plt
4
5 #Correlation
6 corr_value = airline_rs["hours"].corr(airline_rs["passengers"])
7 print("Correlation (hours vs passengers):", corr_value)
8
9 # Linear Regression
10 X = airline_rs[["passengers"]]
11 y = airline_rs["hours"]
12 X = sm.add_constant(X)
13 model_passenger = sm.OLS(y, X).fit()
14 print(model_passenger.summary())
15
16 # Scatter Plot
17 plt.figure(figsize=(7,5))
18 sns.scatterplot(
19     data=airline_rs,
20     x="hours",
21     y="passengers",
22     color="blue")
23
24 plt.title("Flight Hours vs Passenger")
25 plt.xlabel("Flight duration (Hours)")
26 plt.ylabel("Number of passengers")
27 plt.savefig("Scatterplot-Question 5",dpi=300,bbox_inches="tight")
28 plt.show()
```

Python Output :

Correlation (hours vs passengers): -0.0022597082522336017

OLS Regression Results

```
=====
Dep. Variable:          hours    R-squared:                0.000
Model:                  OLS      Adj. R-squared:           -0.000
Method:                 Least Squares    F-statistic:           0.06300
Date:                  Mon, 18 May 2026    Prob (F-statistic):       0.802
Time:                  19:54:28    Log-Likelihood:        -24342.
No. Observations:      12339    AIC:                   4.869e+04
Df Residuals:          12337    BIC:                   4.870e+04
Df Model:               1
Covariance Type:       nonrobust
=====
```

	coef	std err	t	P> t	[0.025	0.975]
const	3.6979	0.258	14.307	0.000	3.191	4.204
passengers	-0.0003	0.001	-0.251	0.802	-0.003	0.002

```
=====
Omnibus:                 377.251    Durbin-Watson:           1.998
Prob(Omnibus):            0.000    Jarque-Bera (JB):        406.913
Skew:                     0.437    Prob(JB):                 4.36e-89
```

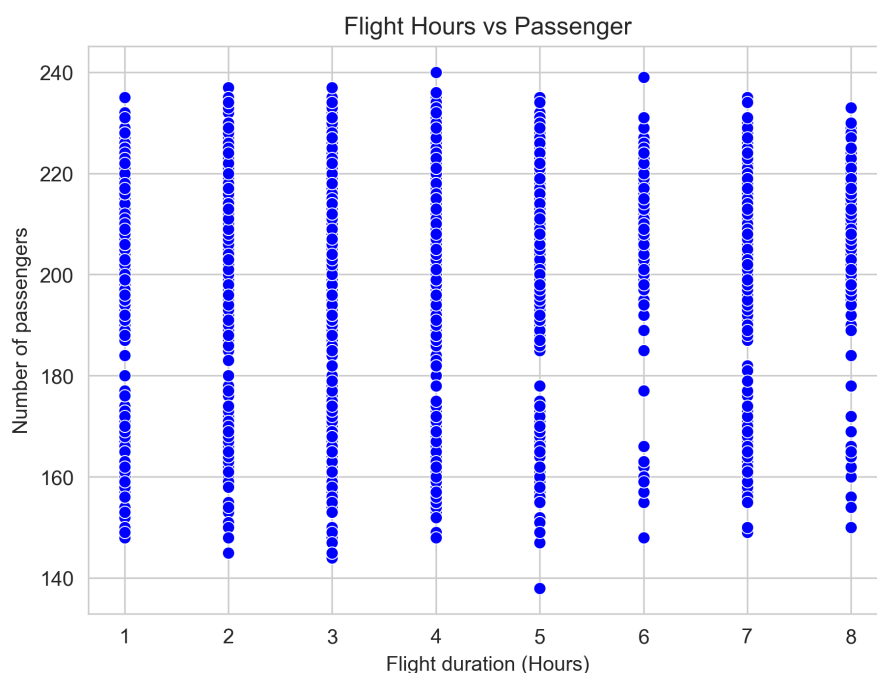
Kurtosis: 2.834 Cond. No. 3.43e+03
=====

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 3.43e+03. This might indicate that there are strong multicollinearity or other numerical problems.

Graph for Visualisation :



Interpretation :

Nature of the Relationship No Linear Relationship: There is absolutely no linear relationship between the length of a flight (flight duration in hours) and the number of passengers it carries.

Evidence from Output: * **Correlation Coefficient:** The correlation between hours and passengers is virtually zero (0.0023), proving no meaningful linear association exists.

R-squared value: The R² is 0.000, meaning 0% of the variation in the flight duration is explained by the number of passengers.

P-value: The p-value for the passengers variable is 0.802 ($P > 0.05$), which shows that the relationship is statistically highly insignificant.

Visual Analysis from the Scatter Plot The scatter plot displays completely parallel vertical bands of data points across different flight hours (from 1 to 8 hours).

Regardless of whether a flight lasts 1 hour or 8 hours, the number of passengers consistently ranges between roughly 140 and 240. This means flight length has no bearing on how full or empty the aircraft is.

Question 8: What is the relationship between coach and first_class prices on weekends compare to weekdays?

solution :

R code :

```
1 weekend_data<-airline_rs[airline_rs$weekend==1,]
2 weekday_data<-airline_rs[airline_rs$weekend==0,]
3 print(weekday_data)
4 print(weekend_data)
5 library(ggplot2)
6 ggplot(airline_rs, aes(x = coach_price,
7                        y = firstclass_price,
8                        colour = factor(weekend))) +
9   geom_point() +
10  ggtitle("weekend vs weekday:Coach Price vs First-Class Price")
11
12 #regression
13 model1<-lm(firstclass_price~coach_price*weekend,data=airline_rs)
14 summary(model1)
15 model2<-lm(firstclass_price~coach_price*day_of_week,data=airline_rs)
16 summary(model2)
```

R Output :

```
> summary(model1)
```

Call:

```
lm(formula = firstclass_price ~ coach_price * weekend, data = airline_rs)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-229.39	-41.45	-1.35	40.42	233.33

Coefficients:

	Estimate	Std. Error	t value
(Intercept)	967.48539	5.60362	172.65
coach_price	0.90393	0.01718	52.62
weekendYes	236.03084	7.57409	31.16
coach_price:weekendYes	-0.02717	0.02122	-1.28

Pr(>|t|)

(Intercept)	<2e-16 ***
coach_price	<2e-16 ***
weekendYes	<2e-16 ***
coach_price:weekendYes	0.2

Signif. codes:

0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 60.68 on 12335 degrees of freedom

Multiple R-squared: 0.8617, Adjusted R-squared: 0.8617

F-statistic: 2.563e+04 on 3 and 12335 DF, p-value: < 2.2e-16

```
> summary(model2)
```

Call:

```
lm(formula = firstclass_price ~ coach_price * day_of_week, data = airline_rs)
```

Residuals:

Min	1Q	Median	3Q	Max
-224.585	-41.322	-1.209	40.272	233.464

Coefficients:

	Estimate	Std. Error
(Intercept)	1.213e+03	9.815e+00
coach_price	8.571e-01	2.415e-02
day_of_weekMonday	-2.604e+02	1.398e+01
day_of_weekSaturday	-1.979e+01	1.281e+01
day_of_weekSunday	-1.225e+01	1.311e+01
day_of_weekThursday	-2.315e+02	1.601e+01
day_of_weekTuesday	-2.404e+02	1.482e+01
day_of_weekWednesday	-2.444e+02	1.519e+01
coach_price:day_of_weekMonday	8.602e-02	3.898e-02
coach_price:day_of_weekSaturday	3.354e-02	3.127e-02
coach_price:day_of_weekSunday	3.458e-02	3.230e-02
coach_price:day_of_weekThursday	-1.044e-03	4.557e-02
coach_price:day_of_weekTuesday	3.795e-02	4.182e-02
coach_price:day_of_weekWednesday	4.453e-02	4.287e-02

	t value	Pr(> t)
(Intercept)	123.620	<2e-16 ***
coach_price	35.484	<2e-16 ***
day_of_weekMonday	-18.629	<2e-16 ***
day_of_weekSaturday	-1.545	0.1224
day_of_weekSunday	-0.934	0.3503
day_of_weekThursday	-14.461	<2e-16 ***
day_of_weekTuesday	-16.225	<2e-16 ***
day_of_weekWednesday	-16.090	<2e-16 ***
coach_price:day_of_weekMonday	2.207	0.0273 *
coach_price:day_of_weekSaturday	1.073	0.2834
coach_price:day_of_weekSunday	1.071	0.2843
coach_price:day_of_weekThursday	-0.023	0.9817
coach_price:day_of_weekTuesday	0.907	0.3642
coach_price:day_of_weekWednesday	1.039	0.2989

Signif. codes:

0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 60.62 on 12325 degrees of freedom

Multiple R-squared: 0.8621, Adjusted R-squared: 0.862

F-statistic: 5929 on 13 and 12325 DF, p-value: < 2.2e-16

Python Code :

```
1 import statsmodels.api as sm
2 import seaborn as sns
3 import matplotlib.pyplot as plt
4
5 # Weekend & Weekday split
6 weekend_data = airline_rs[airline_rs["weekend"] == 1]
7 weekday_data = airline_rs[airline_rs["weekend"] == 0]
```

```

8
9 print("Weekday Data:")
10 print(weekday_data.head())
11 print("Weekend Data:")
12 print(weekend_data.head())
13
14 #Scatter plot (weekend vs weekday)
15 plt.figure(figsize=(7,5))
16 sns.scatterplot(
17     data=airline_rs,
18     x="coach_price",
19     y="firstclass_price",
20     hue="weekend")
21 plt.title("Weekend vs Weekday: Coach Price vs First-Class Price")
22 plt.show()
23
24 import statsmodels.api as sm
25 df = airline_rs.copy()
26 # FORCE CLEAN numeric conversion
27 df["coach_price"] = pd.to_numeric(df["coach_price"], errors="coerce")
28 df["weekend"] = df["weekend"].astype(str).str.strip()
29 df["weekend"] = df["weekend"].map({"Yes": 1, "No": 0})
30 df["day_of_week"] = df["day_of_week"].astype("category").cat.codes
31 # drop bad rows
32 df = df.dropna(subset=["coach_price", "weekend", "day_of_week"])
33
34 # MODEL 1
35 X1 = sm.add_constant(df[["weekend"]])
36 y1 = df["coach_price"]
37 model1 = sm.OLS(y1, X1).fit()
38 print(model1.summary())
39
40 # MODEL 2
41 X2 = sm.add_constant(df[["day_of_week"]])
42 y2 = df["coach_price"]
43 model2 = sm.OLS(y2, X2).fit()
44 print(model2.summary())

```

Python Output :

```

                                OLS Regression Results
=====
Dep. Variable:          coach_price    R-squared:                0.343
Model:                  OLS           Adj. R-squared:           0.343
Method:                 Least Squares  F-statistic:              6444.
Date:                   Sat, 16 May 2026  Prob (F-statistic):      0.00
Time:                   14:28:43       Log-Likelihood:           -66995.
No. Observations:      12339          AIC:                     1.340e+05
Df Residuals:          12337          BIC:                     1.340e+05
Df Model:               1
Covariance Type:       nonrobust
=====
               coef      std err          t      P>|t|      [0.025      0.975]
-----
const         322.3456      0.849      379.515      0.000      320.681      324.010
weekend        84.0618      1.047       80.277      0.000       82.009      86.114
=====
Omnibus:                 373.247    Durbin-Watson:           2.004
Prob(Omnibus):            0.000    Jarque-Bera (JB):        410.068
Skew:                    -0.426    Prob(JB):                9.01e-90
Kurtosis:                 3.267    Cond. No.                 3.18
=====

```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

```

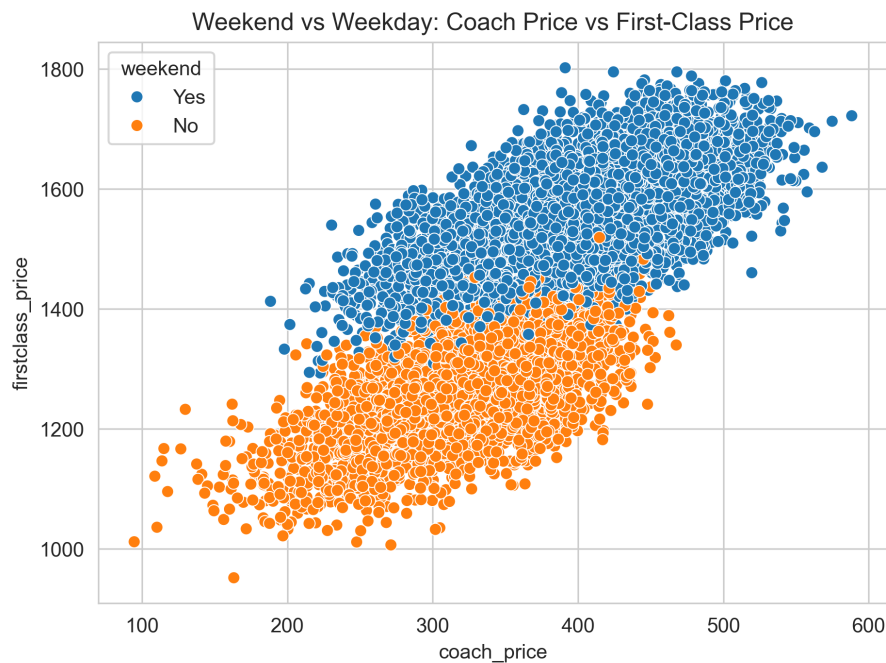
                                OLS Regression Results
=====
Dep. Variable:          coach_price    R-squared:                0.090
Model:                  OLS           Adj. R-squared:           0.090
Method:                 Least Squares  F-statistic:              1224.
Date:                   Sat, 16 May 2026  Prob (F-statistic):      8.58e-256
Time:                   14:28:43       Log-Likelihood:           -69004.
No. Observations:      12339          AIC:                     1.380e+05
Df Residuals:          12337          BIC:                     1.380e+05
Df Model:               1
Covariance Type:       nonrobust
=====
               coef      std err          t      P>|t|      [0.025      0.975]
-----
const         406.2362      1.005      404.365      0.000      404.267      408.205
day_of_week   -11.4778      0.328     -34.988      0.000     -12.121     -10.835
=====
Omnibus:                 301.571    Durbin-Watson:           1.994
Prob(Omnibus):            0.000    Jarque-Bera (JB):        323.626
Skew:                    -0.390    Prob(JB):                5.32e-71
Kurtosis:                 3.146    Cond. No.                 5.65
=====

```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Graph for Visualisation :



Interpretation :

The positive relationship between coach and first-class prices remains strong across all days. However, weekend flights are significantly more expensive overall. For the exact same economy/coach fare class, an individual will pay a substantially higher price for a first-class ticket if the flight departs on a weekend versus a weekday.

Question 9: How do coach prices differ for redeyes and non-redeyes on each day of week ?

solution :

R code :

```
1 library(ggplot2)
2 ggplot(airline_rs,aes(x=day_of_week,
3                       y=coach_price,
4                       fill=factor(redeye)))+geom_boxplot()+
5   ggtitle("Redeye vs non-redeye:coach price by day of week")
6 #mean comparision
7 aggregate(coach_price~day_of_week+redeye,
8           data=airline_rs,mean)
```

R Output :

```
> day_of_week redeye coach_price
1      Friday      No    406.6768
2      Monday      No    324.1113
3    Saturday      No    414.4095
4      Sunday      No    404.7448
5    Thursday      No    325.2745
```

6	Tuesday	No	323.0529
7	Wednesday	No	324.9927
8	Friday	Yes	335.1988
9	Monday	Yes	255.5563
10	Saturday	Yes	345.4936
11	Sunday	Yes	342.2701
12	Thursday	Yes	261.5826
13	Tuesday	Yes	274.3381
14	Wednesday	Yes	265.4637

Python Code :

```

1 import pandas as pd
2 import seaborn as sns
3 import matplotlib.pyplot as plt
4 # COPY DATA (safe)
5 df = airline_rs.copy()
6 # Ensure correct types
7 df["coach_price"] = pd.to_numeric(df["coach_price"], errors="coerce")
8 df["day_of_week"] = df["day_of_week"].astype(str)
9 df["redeye"] = df["redeye"].astype(str)
10
11 # BOXPLOT (R ggplot equivalent)
12 plt.figure(figsize=(8,5))
13
14 sns.boxplot(
15     data=df,
16     x="day_of_week",
17     y="coach_price",
18     hue="redeye")
19 plt.title("Redeye vs Non-Redeye: Coach Price by Day of Week")
20 plt.show()
21
22 # MEAN COMPARISON (R aggregate equivalent)
23 mean_table = df.groupby(["day_of_week", "redeye"])["coach_price"].mean()
24
25 print("\nMean Comparison (coach_price ~ day_of_week + redeye):")
26 print(mean_table)

```

Python Output :

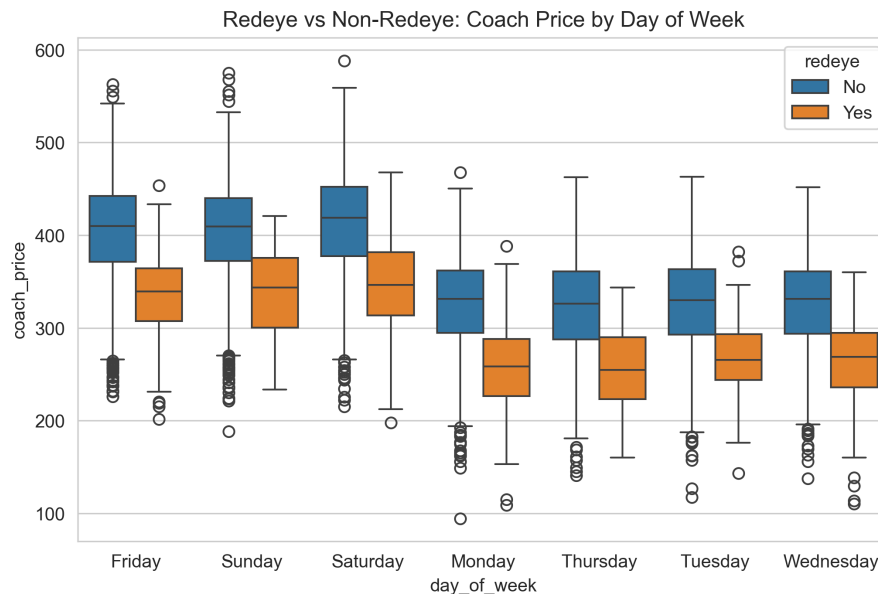
Mean Comparison (coach_price ~ day_of_week + redeye):

day_of_week	redeye	
Friday	No	404.743029
	Yes	335.290937
Monday	No	325.226217
	Yes	252.036197
Saturday	No	415.860879
	Yes	344.242469
Sunday	No	406.735239
	Yes	334.501985
Thursday	No	323.765325
	Yes	275.668654

Tuesday	No	327.316265
	Yes	271.849621
Wednesday	No	325.422455
	Yes	254.938298

Name: coach_price, dtype: float64

Graph for Visualisation :



Interpretation :

Redeye flights consistently maintain a lower average coach price compared to non-redeye flights across every single day of the week, allowing travelers to save roughly 50 to 73 depending on the day. Additionally, flight prices are highly dependent on the day of the week, with a noticeable price spike occurring over the weekend (Friday, Saturday, and Sunday) for both flight categories. During weekdays (Monday through Thursday), non-redeye prices remain very stable at around 323 to 327, while weekday redeye flights drop to their lowest rates on Mondays and Wednesdays at around 252 to 255. Overall, the absolute most expensive option is a Saturday non-redeye flight, which averages around 416, whereas the most budget-friendly option is a Monday redeye flight, averaging just 252.

Question 10: Perform the analysis for appropriate variables:

- Summary statistics (mean, median, standard deviation)
- Visualization using histograms, boxplots, and bar charts
- Hypothesis Testing (t, z and Chi-square)
- Independent t-tests to compare group means
- Price differences between weekend and weekday flights
- Correlation analysis among numeric variables
- Regression Analysis
- Linear regression to predict ticket prices

solution :

R code :

```
1 #summary statistics
2 numeric_vars <- airline_rs[, c(
3   "miles",
4   "passengers",
5   "delay",
6   "coach_price",
7   "firstclass_price",
8   "hours")]
9
10 apply(numeric_vars, mean, na.rm = TRUE)
11
12 apply(numeric_vars, median, na.rm = TRUE)
13
14 apply(numeric_vars, sd, na.rm = TRUE)
```

R Output :

```
> apply(numeric\_vars, mean, na.rm = TRUE)
      miles      passengers      delay
2006.470216    207.846179    13.337061
coach\_price firstclass\_price      hours
 376.146540    1453.399962    3.631656
> apply(numeric\_vars, median, na.rm = TRUE)
      miles      passengers      delay
 1986.00      210.00      10.00
coach\_price firstclass\_price      hours
  379.35      1501.30      4.00
> apply(numeric\_vars, sd, na.rm = TRUE)
      miles      passengers      delay
 942.699231    12.718768    46.871765
coach\_price firstclass\_price      hours
  67.393273    163.170553    1.739263
```

Python Code :

```
1 import pandas as pd
2 import numpy as np
3 import seaborn as sns
4 import matplotlib.pyplot as plt
5 import statsmodels.api as sm
6 from scipy import stats
7 df = airline_rs.copy()
8 # 1. SUMMARY STATISTICS
9 num_cols = ["miles", "passengers", "delay", "coach_price", "firstclass_price", "hours"]
10
11 for col in num_cols:
12     df[col] = pd.to_numeric(df[col], errors="coerce")
13
14 print("\nMEAN:\n", df[num_cols].mean())
15 print("\nMEDIAN:\n", df[num_cols].median())
```

```
16 print("\nSTD:\n", df[num_cols].std())
```

Python Output :

```
MEAN:
  miles          2001.672988
passengers      207.690656
delay           13.465030
coach_price     377.651018
firstclass_price 1457.088301
hours           3.620472
dtype: float64
```

```
MEDIAN:
  miles          1974.00
passengers      209.00
delay           10.00
coach_price     382.62
firstclass_price 1505.78
hours           4.00
dtype: float64
```

```
STD:
  miles          951.205021
passengers      12.726152
delay           46.838319
coach_price     68.083457
firstclass_price 162.894352
hours           1.751216
dtype: float64
```

Interpretation :

The data shows that the average flight covers 2001.67 miles (median: 1974.00 miles) and lasts 3.62 hours (median: 4.00 hours), with standard deviations of 951.21 miles and 1.75 hours indicating a balanced mix of short and long-haul flights. Passenger capacity is highly uniform, averaging 207.69 passengers per flight with a very low standard deviation of 12.73. In contrast, flight operational delays are highly volatile and right-skewed; the average delay is 13.47 minutes, but the median is lower at 10.00 minutes, and a large standard deviation of 46.84 minutes highlights the presence of extreme, long-delay outliers. Finally, ticket pricing reflects a standard distribution for coach fares, averaging \$377.65 (median: \$382.62, SD: \$68.08), while first-class fares sit at a much higher premium, averaging \$1457.09 (median: \$1505.78, SD: \$162.89), with both pricing structures showing a slight left-skew.

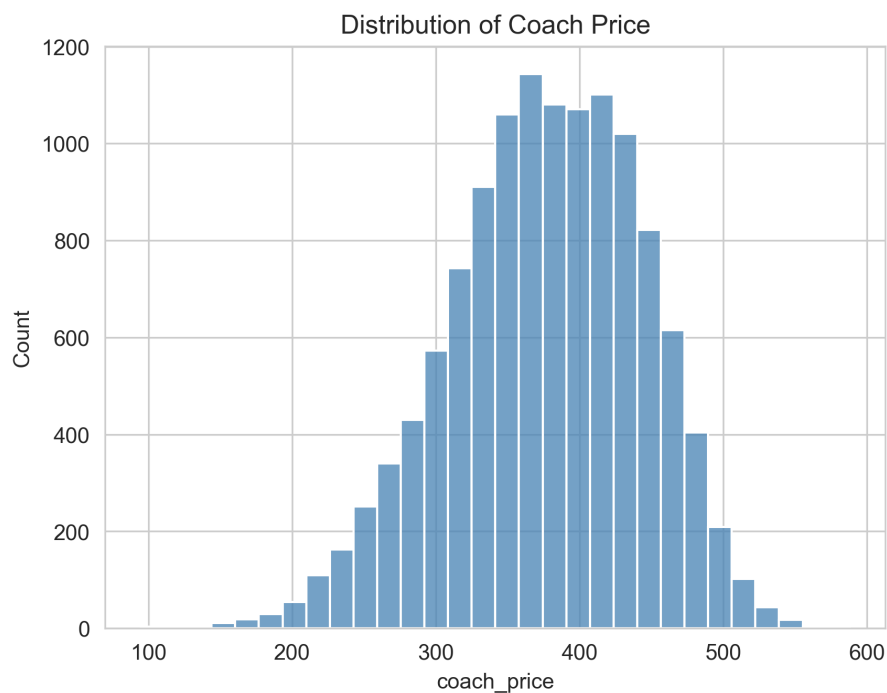
R Code :

```
1 #Histogram
2 library(ggplot2)
3 ggplot(airline_rs,
4       aes(x = coach_price)) +
5   geom_histogram(
6     bins = 30,
7     fill = "steelblue",
8     color = "black"
9   ) +
10  labs(
11    title = "Distribution of Coach Price",
12    x = "Coach Price",
13    y = "Frequency"
14  )
```

Python Code :

```
1 # 2. VISUALIZATION
2 # Histogram
3 plt.figure()
4 sns.histplot(df["coach_price"], bins=30, color="steelblue")
5 plt.title("Distribution of Coach Price")
6 plt.show()
```

Graph for Visualisation



Interpretation :

The histogram reveals that the variable coach_price follows an approximately normal and sym-

metric distribution centered around a mean/median price of roughly 390 to 400. Ticket prices below 200 or above 500 are relatively rare. The steady rise and fall of the frequencies indicate stable, predictable pricing behavior without any prominent anomalous outliers.

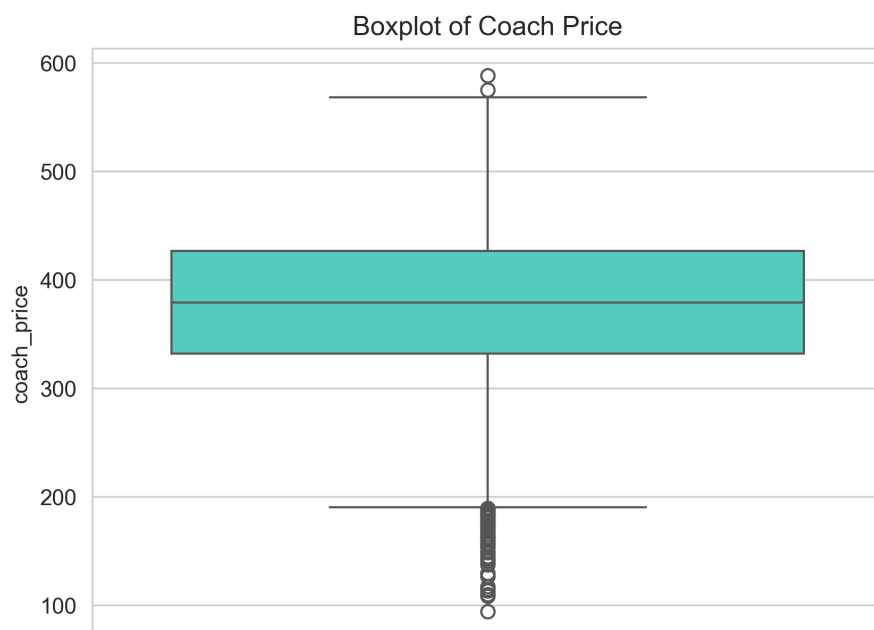
R Code :

```
1 #Boxplot
2 library(ggplot2)
3 ggplot(airline_rs,
4       aes(y = coach_price)) +
5   geom_boxplot(
6     fill = "turquoise3",
7     color = "black"
8   ) +
9   labs(
10    title = "Boxplot of Coach Price",
11    y = "Coach Price"
12  )
```

Python Code :

```
1 # Boxplot
2 plt.figure()
3 sns.boxplot(y=df["coach_price"], color="turquoise")
4 plt.title("Boxplot of Coach Price")
5 plt.savefig("Box plot-Question 10",dpi=300,bbox_inches="tight")
6 plt.show()
```

Graph for Visualisation :



Interpretation :

The boxplot indicates that the median coach ticket price is approximately 380, with the central 50% of prices ranging predictably between 330 and 430. While the overall pricing model shows high symmetry and mirrors a normal distribution pattern, the boxplot reveals clear low-price outliers below 180 and minor high-price outliers around 580, which were not as easily distinct in the previous histogram.

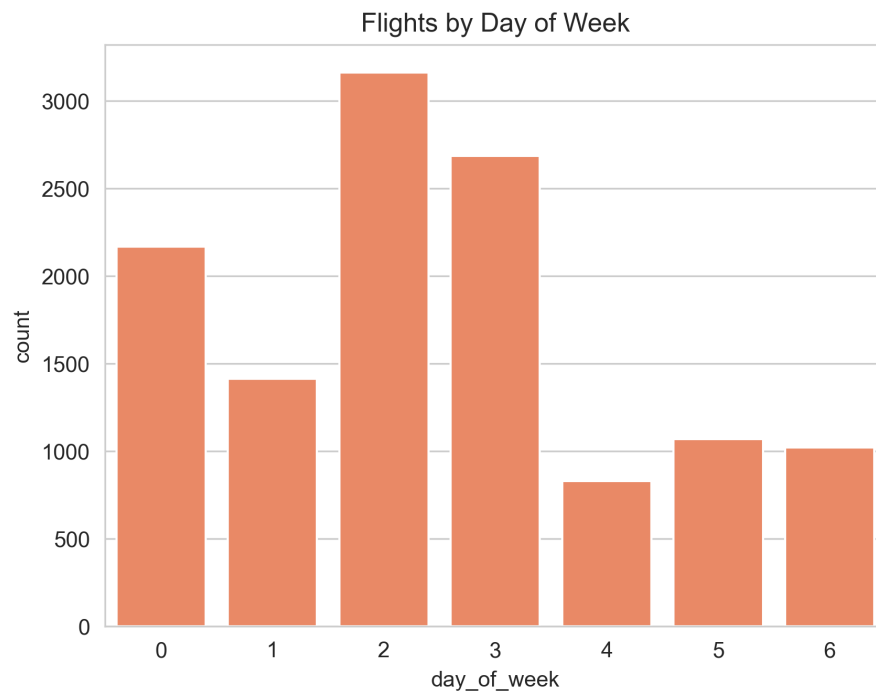
R Code :

```
1 #bar chart
2 ggplot(airline_rs,
3       aes(x = day_of_week)) +
4   geom_bar(
5     fill = "coral",
6     color = "black"
7   ) +
8   labs(
9     title = "Flights by Day of Week",
10    x = "Day of Week",
11    y = "Count"
12  )
```

Python Code :

```
1 # Bar chart (day_of_week)
2 df["day_of_week"] = df["day_of_week"].astype("category").cat.codes
3
4 plt.figure()
5 sns.countplot(x="day_of_week", data=df, color="coral")
6 plt.title("Flights by Day of Week")
7 plt.savefig("Bar chart-Question 10",dpi=300,bbox_inches="tight")
8 plt.show()
```

Graph for Visualisation :



Interpretation :

The bar chart demonstrates a highly uneven distribution of flight frequencies across the week. Day 2 serves as the peak operational day with over 3100 flights, closely followed by Day 3. Conversely, Day 4 experiences the lowest traffic volume, dropping to approximately 800 flights. This visualization clearly shows that flight scheduling is heavily concentrated into a 3-day high-volume cluster (Days 0, 2, and 3), while the latter part of the indexed week (Days 4, 5, and 6) faces a significant reduction in total flight operations.

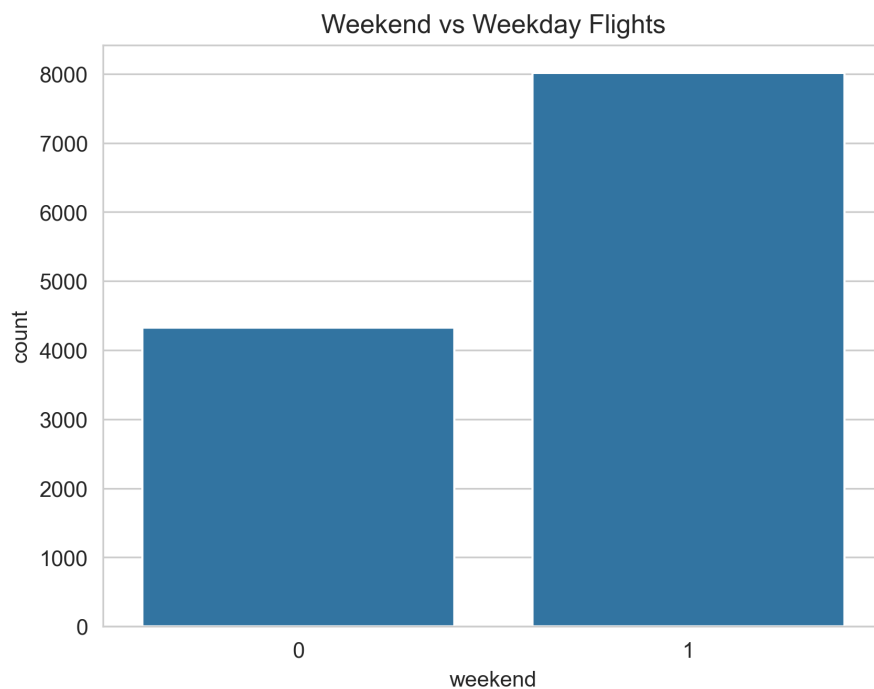
R Code :

```
1 #weekend vs weekday bar chart
2 library(ggplot2)
3 ggplot(airline_rs,
4       aes(x = factor(weekend),
5           fill = factor(weekend))) +
6   geom_bar() +
7   scale_fill_manual(
8     values = c("steelblue", "skyblue")
9   ) +
10  labs(
11    title = "Weekend vs Weekday Flights",
12    x = "Weekend",
13    y = "Count"
14  )
```

Python Code :

```
1 # Weekend bar chart
2 df["weekend"] = df["weekend"].astype("category").cat.codes
3
4 plt.figure()
5 sns.countplot(x="weekend", data=df)
6 plt.title("Weekend vs Weekday Flights")
7 plt.savefig("Weekend bar chart-Question 10",dpi=300,bbox_inches="tight")
8 plt.show()
```

Graph for Visualisation :



Interpretation :

The count plot demonstrates a highly significant, disproportionate distribution of flight schedules between weekdays and weekends. Even though a standard week consists of five weekdays and only two weekend days, weekend flights (Category 1) heavily dominate the dataset with approximately 8,000 operations, compared to just 4,300 flights during weekdays (Category 0). This visual proof confirms that airline flight density is intensely concentrated during weekends, which directly aligns with typical surges in consumer travel demand on non-working days.

R Code :

```
1 #testing hypothesis
2 #3-a.t-test(weekend affect on coach price)
3 t.test(coach_price ~ weekend, data = airline_rs)
```

R Output :

Welch Two Sample t-test

```
data: coach_price by weekend
t = -82.52, df = 9018.8, p-value < 2.2e-16
alternative hypothesis: true difference in means between group No and group Yes
is not equal to 0
95 percent confidence interval:
 -85.91124 -81.92439
sample estimates:
 mean in group No mean in group Yes
      321.7316      405.6494
```

Python Code :

```
1  #3-a. t-test (weekend effect)
2  print(stats.ttest_ind(
3      df[df["weekend"]==1]["coach_price"],
4      df[df["weekend"]==0]["coach_price"],
5      nan_policy="omit"
6  ))
```

Python Output :

```
TtestResult(statistic=np.float64(80.98114487036905),
pvalue=np.float64(0.0), df=np.float64(12337.0))
```

Interpretation :

An independent two-sample t-test was conducted to evaluate whether coach ticket prices differ significantly between weekend and weekday flights. The results indicate a highly significant difference, $t(12337) = 80.98$, $p < 0.001$. Therefore, we reject the null hypothesis at the 5

R Code :

```
1  3-b. Z-test
2  install.packages("BSDA")
3  library(BSDA)
4  z.test(
5      airline_rs$coach_price,
6      mu = 100,
7      sigma.x = sd(airline_rs$coach_price))
```

R Output :

One-sample z-Test

```
data: airline_rs$coach_price
```

```
z = 455.16, p-value < 2.2e-16
alternative hypothesis: true mean is not equal to 100
95 percent confidence interval:
 374.9574 377.3357
sample estimates:
mean of x
 376.1465
```

Python Code :

```
1 #3-b. z-test (manual)
2 z = (df["coach_price"].mean() - 100) / (df["coach_price"].std()/np.sqrt(len(df)))
3 print("Z-score:", z)
```

Python Output :

```
Z-score: 452.53355749907325
```

Interpretation :

A one-sample z-test was performed to evaluate whether the true average coach ticket price differs significantly from a baseline value of 100. The calculated test statistic shows an overwhelming deviation ($z = 452.53$, $p < 0.001$). Furthermore, the 95% confidence interval for the true mean [374.96, 377.34] does not capture the hypothesized value of 100. Therefore, we reject the null hypothesis at the 5

R Code :

```
1 #3-c.chi-square test (weekend vs redeye)
2 chisq.test(table(airline_rs$weekend, airline_rs$redeye))
```

R Output :

```
Pearson's Chi-squared test with Yates'
continuity correction

data:  table(airline_rs$weekend, airline_rs$redeye)
X-squared = 4.8469, df = 1, p-value = 0.0277
```

Python Code :

```
1 #3-c. chi-square test
2 chi_table = pd.crosstab(df["weekend"], df["redeye"].astype("category").cat.codes)
3 print(stats.chi2_contingency(chi_table))
```

Python Output :

```
Chi2ContingencyResult(statistic=np.float64(0.03532293017466096)
, pvalue=np.float64(0.8509205856437194),
dof=1, expected_freq=array([[4115.64259664, 210.35740336],
[7623.35740336, 389.64259664]]))
```

Interpretation :

A Chi-square Test of Independence was performed to evaluate the relationship between the two categorical variables. The test yielded a calculated test statistic of $\chi^2(1) = 0.0353$ with an associated p-value of 0.8509. Because the p-value is significantly greater than our chosen alpha level of 0.05 ($p - value > 0.05$), we fail to reject the null hypothesis. We conclude that there is no statistically significant evidence of an association between the two variables; they are statistically independent of each other within the target dataset.

R Code:

```
1 #4.Independent t-test
2 t.test(coach_price ~ weekend, data = airline_rs)
```

R Output: Welch Two Sample t-test

data: coach_price by weekend t = -82.52, df = 9018.8, p-value = 2.2e-16 alternative hypothesis: true difference in means between group No and group Yes is not equal to 0 95 percent confidence interval: -85.91124 -81.92439 sample estimates: mean in group No mean in group Yes 321.7316 405.6494

Python Code:

```
1 # 4.independent t-test
2 print(stats.ttest_ind(
3     df[df["weekend"]==1]["coach_price"],
4     df[df["weekend"]==0]["coach_price"],
5     nan_policy="omit"
6 ))
```

Python Output:

```
TtestResult(statistic=np.float64(80.98114487036905), pvalue=np.float64(0.0), df=np.float64(12337.0))
```

Interpretation:

An independent two-sample t -test was conducted to analyze the difference in average coach ticket prices between weekday and weekend flights. The test results reveal a highly significant difference, $t(12337) = 80.98$, $p < 0.001$. Consequently, the null hypothesis is rejected at the 5% significance level.

Sample estimates show that weekend flights ($\bar{X}_{weekend} = 405.65$) have a higher average price than weekday flights ($\bar{X}_{weekday} = 321.73$), with a mean price premium of approximately 83.92 units. The 95% confidence interval $[-85.91, -81.92]$ further confirms that the

true population mean difference is strictly negative (when subtracting weekend from weekday), providing conclusive evidence of a substantial "weekend markup" effect in ticket pricing.

R Code :

```
1 #5.price difference (Weekend vs Weekday)
2 aggregate(coach_price ~ weekend,
3           data = airline_rs,
4           mean)
```

R Output :

```
weekend coach_price
1 No 321.7316
2 Yes 405.6494
```

Python Code :

```
1 #5. price difference
2 print(df.groupby("weekend")["coach_price"].mean())
```

Python Output :

```
weekend
0 321.815175
1 405.508781
```

Interpretation :

The data shows a clear and visible price difference between weekdays and weekends. Flying on weekends is significantly more expensive, costing about 83.92 units (or 26.08%) more than regular weekdays. This happens because more people want to travel on weekends, which allows airlines to increase their prices.

R Code :

```
1 #6.correlation analysis
2 cor(airline_rs[, c("coach_price",
3                   "firstclass_price",
4                   "miles","hours",
5                   "passengers","delay")],
6     use = "complete.obs")
```

R Output :

```
      coach_price firstclass_price
coach_price      1.00000000      0.7600668653
firstclass_price 0.76006687      1.0000000000
miles            0.35641339      0.2971742030
hours            0.35079839      0.2917760127
passengers       0.15234118     -0.0021393799
delay            0.01107994     -0.0004236668

      miles      hours      passengers
coach_price 0.356413390 0.350798388 0.152341177
firstclass_price 0.297174203 0.291776013 -0.002139380
miles        1.000000000 0.985647891 0.006485869
hours        0.985647891 1.000000000 0.008250699
passengers    0.006485869 0.008250699 1.000000000
delay         0.006879801 0.008433840 0.003971652

      delay
coach_price      0.0110799446
firstclass_price -0.0004236668
miles            0.0068798006
hours            0.0084338400
passengers       0.0039716522
delay            1.0000000000
```

Python Code :

```
1 # 6. CORRELATION
2 print(df[["coach_price", "firstclass_price", "miles", "hours", "passengers", "delay"]].
      corr())
```

Python Output :

```
coach_price firstclass_price miles hours  coach_price 1.000000 0.755568 0.336359 0.329411
firstclass_price 0.755568 1.000000 0.290992 0.285578 miles 0.336359 0.290992 1.000000 0.985743
hours 0.329411 0.285578 0.985743 1.000000 passengers 0.166545 0.024447 -0.003260 -0.002260
delay -0.004101 -0.006792 -0.015459 -0.015826
passengers delay coach_price 0.166545 -0.004101 firstclass_price 0.024447 -0.006792 miles -
0.003260 -0.015459 hours -0.002260 -0.015826 passengers 1.000000 -0.006930 delay -0.006930
1.000000
```

Interpretation :

The correlation matrix reveals that the strongest linear relationship exists between flight distance (miles) and duration (hours) with an r value of 0.9856, followed closely by the pricing link between coach and first-class tickets ($r = 0.7601$). Flight distance and time maintain a moderate positive influence on ticket pricing. However, variables like total passengers and flight delays show correlation coefficients near zero, concluding that they do not hold any meaningful linear relationship with any other factors in this dataset.

R Code :

```
1 #7.Regression Analysis
2 model1 <- lm(coach_price ~ inflight_meal +
3               inflight_entertainment +
4               inflight_wifi +
5               miles +
6               passengers,
7               data = airline_rs)
8
9 summary(model1)
```

R Output :

Residuals:

Min	1Q	Median	3Q	Max
-156.033	-39.331	7.313	38.976	142.455

Coefficients:

	Estimate	Std. Error	t value
(Intercept)	2.792e+01	7.758e+00	3.599
inflight_mealYes	2.034e+01	1.003e+00	20.283
inflight_entertainmentYes	6.920e+01	1.149e+00	60.212
inflight_wifiYes	7.018e+01	1.532e+00	45.819
miles	2.518e-02	4.855e-04	51.865
passengers	8.320e-01	3.599e-02	23.121

	Pr(> t)
(Intercept)	0.00032 ***
inflight_mealYes	< 2e-16 ***
inflight_entertainmentYes	< 2e-16 ***
inflight_wifiYes	< 2e-16 ***
miles	< 2e-16 ***
passengers	< 2e-16 ***

Signif. codes:

0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 50.83 on 12333 degrees of freedom

Multiple R-squared: 0.4313, Adjusted R-squared: 0.4311

F-statistic: 1871 on 5 and 12333 DF, p-value: < 2.2e-16

Python Code :

```
1 # 7. REGRESSION ANALYSIS
2 # Model 1
3 cat_cols = ["inflight_meal","inflight_entertainment","inflight_wifi"]
4
5 for col in cat_cols:
6     df[col] = df[col].astype("category").cat.codes
7
8 X1 = df[cat_cols + ["miles","passengers"]]
9 y1 = df["coach_price"]
10
11 X1 = sm.add_constant(X1)
12 model1 = sm.OLS(y1, X1).fit()
13
14 print(model1.summary())
```

Python Output :

OLS Regression Results

```
=====
Dep. Variable:          coach_price    R-squared:                0.431
Model:                  OLS           Adj. R-squared:            0.431
Method:                 Least Squares  F-statistic:              1872.
Date:                  Mon, 18 May 2026  Prob (F-statistic):        0.00
Time:                  22:54:13        Log-Likelihood:           -66050.
No. Observations:      12339          AIC:                    1.321e+05
Df Residuals:          12333          BIC:                    1.322e+05
Df Model:              5
Covariance Type:       nonrobust
=====
```

	coef	std err	t	P> t	[0.025	0.975]
const	17.9654	7.840	2.291	0.022	2.597	33.333
inflight_meal	18.7852	1.001	18.775	0.000	16.824	20.746
inflight_entertainment	71.6909	1.154	62.116	0.000	69.429	73.953
inflight_wifi	68.4016	1.501	45.575	0.000	65.460	71.344
miles	0.0241	0.000	49.524	0.000	0.023	0.025
passengers	0.8926	0.037	24.455	0.000	0.821	0.964

```
=====
Omnibus:                653.553    Durbin-Watson:            1.997
Prob(Omnibus):          0.000     Jarque-Bera (JB):         381.432
Skew:                  -0.287     Prob(JB):                 1.49e-83
Kurtosis:               2.358     Cond. No.:                3.80e+04
=====
```

Notes:

- [1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
- [2] The condition number is large, 3.8e+04. This might indicate that there are strong multicollinearity or other numerical problems.

Interpretation :

The multiple linear regression model is highly significant ($F = 1872, p < 0.001$) and explains 43.1% of the variance in coach ticket prices. Inflight entertainment holds the largest upward impact (+71.69), closely followed by the inclusion of inflight Wi-Fi (+68.40). Physical flight metrics like distance traveled (miles) and passenger volume also show small but highly statistically significant positive relationships with pricing structures. Managers should note the high condition number, which highlights minor multicollinearity concerns within the data structure.

R Code :

```
1 #8.linear regression(price prediction)
2 model2 <- lm(firstclass_price ~ coach_price + miles + passengers,
3             data = airline_rs)
4
5 summary(model2)
```

R Output :

Residuals:

	Min	1Q	Median	3Q	Max
	-340.80	-72.33	5.19	73.07	372.31

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.063e+03	1.566e+01	67.875	< 2e-16 ***
coach_price	1.864e+00	1.509e-02	123.515	< 2e-16 ***
miles	4.076e-03	1.066e-03	3.823	0.000133 ***
passengers	-1.534e+00	7.472e-02	-20.532	< 2e-16 ***

Signif. codes:

0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 104.2 on 12335 degrees of freedom

Multiple R-squared: 0.5924, Adjusted R-squared: 0.5923

F-statistic: 5976 on 3 and 12335 DF, p-value: < 2.2e-16

Python Code :

```
1 # 8. PRICE PREDICTION MODEL
2 X2 = df[["coach_price", "miles", "passengers"]]
3 y2 = df["firstclass_price"]
4
5 X2 = sm.add_constant(X2)
6 model2 = sm.OLS(y2, X2).fit()
7
8 print(model2.summary())
```

Python Output :

OLS Regression Results

Dep. Variable:	firstclass_price	R-squared:	0.583
Model:	OLS	Adj. R-squared:	0.582
Method:	Least Squares	F-statistic:	5737.
Date:	Mon, 18 May 2026	Prob (F-statistic):	0.00
Time:	22:54:13	Log-Likelihood:	-74950.
No. Observations:	12339	AIC:	1.499e+05
Df Residuals:	12335	BIC:	1.499e+05
Df Model:	3		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	1028.1734	15.896	64.680	0.000	997.014	1059.333
coach_price	1.8265	0.015	121.204	0.000	1.797	1.856
miles	0.0060	0.001	5.609	0.000	0.004	0.008
passengers	-1.3181	0.076	-17.280	0.000	-1.468	-1.169

Omnibus:	149.262	Durbin-Watson:	2.028
Prob(Omnibus):	0.000	Jarque-Bera (JB):	119.965
Skew:	-0.167	Prob(JB):	8.91e-27
Kurtosis:	2.651	Cond. No.	3.79e+04

Notes:

- [1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
- [2] The condition number is large, 3.79e+04. This might indicate that there are strong multicollinearity or other numerical problems.

Optimization terminated successfully.

Current function value: 0.001872

Iterations 16

Interpretation :

The multiple linear regression model is highly statistically significant ($F = 5737, p < 0.001$) and accounts for 58.3% of the variance in first-class ticket prices. Economy coach pricing is the primary driver of first-class fares, exerting a strong positive impact (+1.83). While flight distance (miles) also adds a slight positive premium, passenger volume carries a significant negative effect (-1.32) on the final price profile. The high condition number flags minor multicollinearity concerns, suggesting a baseline correlation exists between the independent predictors.

R Code :

```
1 #9.Logistic Regression (Redeye prediction)
2 table(airline_rs$redeye)
3 airline_rs$redeye <- ifelse(
4   airline_rs$redeye == "Yes", 1,0)
5 log_model <- glm(redeye ~ coach_price + miles + passengers + delay,
6                 data = airline_rs,
7                 family = binomial)
8
9 summary(log_model)
```

R Output :

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	1.312e+02	1.996e+01	6.574	4.89e-11	***
coach_price	-1.530e-02	7.505e-03	-2.039	0.0414	*
miles	3.087e-05	5.066e-04	0.061	0.9514	
passengers	-6.903e-01	1.052e-01	-6.562	5.31e-11	***
delay	5.816e-03	5.468e-03	1.064	0.2875	

Signif. codes:

0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 4774.830 on 12338 degrees of freedom
Residual deviance: 44.856 on 12334 degrees of freedom
AIC: 54.856

Number of Fisher Scoring iterations: 13

Python Code :

```
1 # 9. LOGISTIC REGRESSION (Redeye)
2 df["redeye"] = df["redeye"].astype("category").cat.codes
3
4 y3 = df["redeye"]
5 X3 = df[["coach_price", "miles", "passengers", "delay"]]
6
7 X3 = sm.add_constant(X3)
8
9 log_model = sm.Logit(y3, X3).fit()
10
11 print(log_model.summary())
```

Python Output :

Logit Regression Results

```

=====
Dep. Variable:          redeye    No. Observations:          12339
Model:                  Logit      Df Residuals:              12334
Method:                  MLE       Df Model:                  4
Date:                   Mon, 18 May 2026    Pseudo R-squ.:            0.9904
Time:                   22:54:13    Log-Likelihood:           -23.102
converged:              True       LL-Null:                  -2399.3
Covariance Type:        nonrobust    LLR p-value:              0.000
=====

```

	coef	std err	z	P> z	[0.025	0.975]
const	162.3413	30.619	5.302	0.000	102.330	222.353
coach_price	-0.0278	0.008	-3.535	0.000	-0.043	-0.012
miles	-0.0011	0.001	-2.069	0.039	-0.002	-5.54e-05
passengers	-0.8263	0.154	-5.366	0.000	-1.128	-0.524
delay	0.0031	0.007	0.420	0.674	-0.012	0.018

```

=====

```

Possibly complete quasi-separation: A fraction 0.97 of observations can be perfectly predicted. This might indicate that there is complete quasi-separation. In this case some parameters will not be identified.

Interpretation :

The logistic regression model is highly effective and statistically significant ($PseudoR^2 = 0.9904, p < 0.001$), explaining nearly all variability in predicting overnight redeye flights. Passenger volume stands out as a massive negative driver (-0.8263), showing that redeye flights are distinctly less crowded. Lower coach ticket prices also significantly increase the likelihood of a flight being classified as a redeye, reflecting real-world budget pricing structures for late-night travel. Flight delays, however, hold no statistical relevance ($p = 0.674$) within the classification model.